May 14, 2026

Grant Review Committee
VTPEH 6182 Public Health Practice - Planning; Request for Proposals
Cornell University, Ithaca, New York

Dear Members of the Grant Review Committee,

We are pleased to submit this proposal on behalf of Group 3B in response to the Planning 6182 Request for Proposals. Our project, the TickSafe Finger Lakes Initiative, is a one-year community-based pilot program designed to reduce tick-borne disease (TBD) incidence in the Finger Lakes region of New York State through active, theory-driven educational outreach.

Tick-borne diseases are a rapidly escalating public health threat in New York State. In 2024, 18,704 confirmed or probable Lyme disease cases were reported statewide; more than a fourfold increase in incidence since 2020. The Finger Lakes region is classified by the New York State Department of Health as a very high-risk area for encountering infected ticks. Despite this, current prevention efforts largely rely on passive information dissemination, which evidence demonstrates is insufficient to drive sustained protective behavior change at the population level.

The TickSafe Finger Lakes Initiative directly addresses this gap. Grounded in Social Cognitive Theory, our program targets three interlocking determinants of behavior: personal knowledge of tick-borne disease risk, environmental visibility of prevention messaging, and behavioral skills for consistent protective action. We pursue three coordinated objectives: (1) expand access to updated prevention education; (2) conduct active community outreach in high-exposure settings; and (3) build sustainable regional partnerships among public health agencies, healthcare providers, and community organizations. This initiative aligns with the RFP's priorities of Sustainability, Equity, and Resilience.

We are requesting $14,313 in grant funding, complemented by $5,600 in in-kind contributions, for a total project cost of $19,913; within the $20,000 ceiling. Our team; Asma Darwish, Hana Bauta, and Leslie Flora, MPH students at Cornell University; bring expertise in public health program planning, health communications, and community engagement. We have established partnerships with Cornell Cooperative Extension, Cornell Integrated Pest Management, Cayuga Health, and local county health departments.

The TickSafe Finger Lakes Initiative is designed not only for regional impact, but as a replicable, evidence-generating pilot that can support future grant-based funding as tick ranges continue to expand northward. We are grateful for the committee's time and consideration.

Respectfully submitted,

Asma Darwish, Hana Bauta, and Leslie Flora
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TickSafe Finger Lakes Initiative

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Tick-borne diseases (TBDs), particularly Lyme disease, are escalating rapidly across New York State. Incidence rose from 37.2 cases per 100,000 residents in 2020 to 164.8 per 100,000 in 2024, with 18,704 confirmed or probable cases reported statewide in 2024 alone. The Finger Lakes region; classified by the New York State Department of Health as very high risk for infected tick encounters; faces compounding vulnerability: forested landscapes, agricultural land use, abundant outdoor recreation, and structural barriers including transportation gaps and limited rural healthcare access. High-risk populations include children aged 0-10; adults aged 40-60, outdoor workers, pet owners, and rural residents.

Current prevention efforts in the region rely primarily on passive information dissemination; surveillance reporting, static educational campaigns, and online resources. Evidence shows that passive dissemination alone is insufficient to change protective behaviors at the population level. Only 39% of community members report ever seeing tick-prevention educational materials, yet those who do are significantly more likely to perform tick checks (75% vs. 64%). This gap between available prevention resources and community adoption of protective behaviors is the central problem this initiative addresses.

The TickSafe Finger Lakes Initiative is a one-year, community-based pilot program grounded in Social Cognitive Theory. It pursues three objectives: (1) develop and disseminate updated, visually accessible tick-prevention educational materials building on existing NYS and Cornell IPM resources; (2) conduct active community outreach; workshops, tabling events, and social media campaigns; in high-exposure settings including schools, parks, libraries, trailheads, and farmers markets; and (3) strengthen regional partnerships among health departments, healthcare providers, Cornell Cooperative Extension, and community organizations to build sustainable prevention capacity beyond the pilot year.

The initiative requests $14,313 in grant funding ($19,913 total with in-kind), for a 12-month project starting July 1. Outcomes will be measured through pre/post participant surveys, event attendance, materials distributed, partnership tracking, and social media metrics. By transitioning from passive to active prevention outreach, TickSafe aims to increase knowledge, improve risk perception, build consistent protective behaviors, and ultimately reduce tick-borne disease incidence in the Finger Lakes region; while also generate evidence to support future, larger-scale funding.

Proposal Narrative

Background

Tick-Borne Disease as an Emerging Public Health Threat

An ever-growing concern for public health professionals is the rising prevalence and geographic expansion of vector-borne diseases into new regions.[1] Public health professionals are responsible not only for conducting surveillance and monitoring the spread of these diseases, but also for providing clear and accessible information to the public about the risks they pose. Vector-borne diseases are infections transmitted to humans through the bite of infected organisms, such as mosquitoes or ticks.[1] These vectors carry viruses, bacteria or parasites that can be transmitted to humans during the feeding process.[1] Notable examples of vector-borne diseases include malaria, transmitted by mosquitoes, and Lyme Disease, transmitted through ticks vectors.[2] Without timely prevention and treatment, vector-borne infections can lead to serious health consequences and place growing strain on public health systems since vector-borne diseases account for more than 700,000 deaths globally each year.[1]

The increasing spread of VBDs further underscores the importance of effective prevention and early detection strategies. The geographic range disease-transmitting vectors is expanding due to a variety of factors, including climate change.[1,3,4] As temperatures increase in regions farther from the equator, environments that were once unsuitable for ticks are becoming more favorable for their survival and reproduction, allowing ticks to expand their geographic range.[3,4] This expansion has enabled ticks to migrate and establish themselves in new ecosystems, contributing to increased prevalence of tick-borne diseases.[3,4] For example, studies have documented a steady northward expansion of the blacklegged *(Ixodes scapularis)* across North America, with populations becoming established in previously low-risk areas as environmental conditions become more suitable.[4,8] Additionally, alterations to habitat suitability for ticks (e.g., warmer winters, longer growing seasons, shifts in vegetation) can increase opportunities for human-vector interactions and therefore raise population-level risk.[3,4] As a result, communities that historically experienced minimal exposure to ticks and mosquitoes while outdoors are now reporting more frequent tick encounters, or worse yet, are unaware of exposure-related risks. Without a greater focus on prevention, it can be difficult to effectively communicate the risks associated with the growing prevalence of tick-borne diseases.

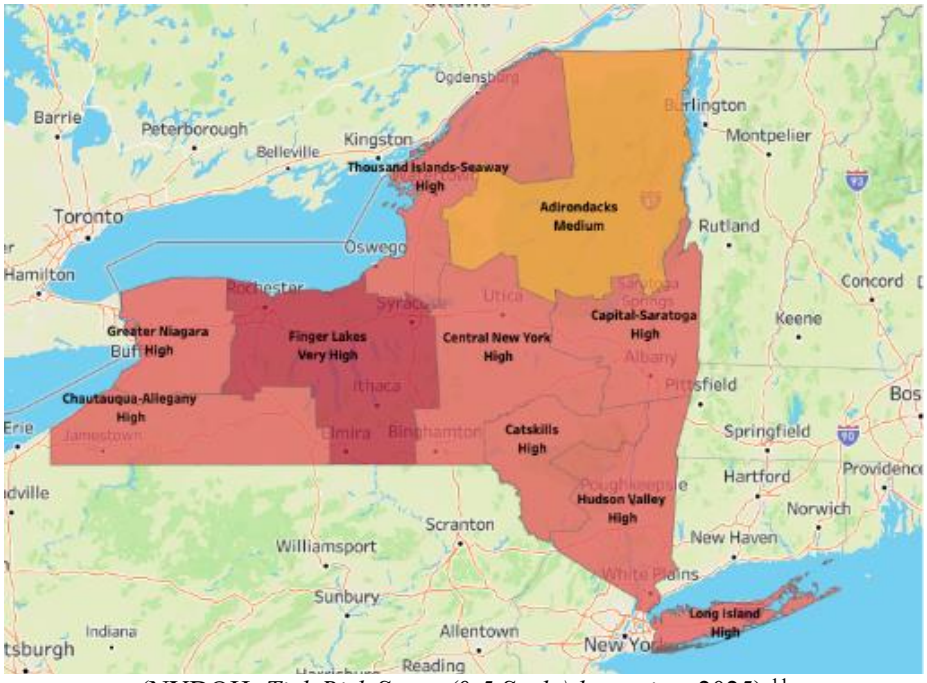
The Disease Burden in New York State

Tick-borne diseases represent a growing public health threat in New York State (NYS).[5-7] Over the past several decades, Lyme disease incidence has expanded geographically from historically high-incidence areas such as the Hudson Valley into central and northern counties of NYS.[5] Historically, southeastern NYS accounted for the majority of Lyme disease cases due to sustained endemic transmission and high tick density.[5,6] However, surveillance data now show that Lyme disease cases are reported in all 57 counties outside of New York City, indicating statewide establishment of *Ixodes scapularis*– commonly known as the blacklegged tick and primary vector responsible for transmitting Lyme disease in the northeastern United States.[5,7] In 2020, Lyme disease incidence in New York State excluding New York City was 37.2 cases per 100,000 residents, and by 2024 it had risen to 164.8 cases per 100,000 residents.[6] In 2024, a total of 18,704 confirmed or probable Lyme disease cases were reported in New York State alone.[6] Certain populations are disproportionately affected, with the highest incidence observed among children aged 5-14 years and adults aged 45-64 years, as well as among males, likely due to increased occupational and recreational outdoor exposure.[15,16] Additionally, the highest case rates continue to be reported in rural and suburban regions of southeastern and central New York, where environmental conditions support higher tick density and human-tick interactions..[5,6]

Start a new para: As seen globally, climate-driven ecological changes such as milder winters, extended spring seasons and increased habitat suitability is an issue of concern within the state, driving the enhanced

tick survival and reproduction process- which is consequently influencing tick populations to expand northward.[5,8,9] Ecological surveillance, which involves monitoring tick populations and pathogen presence in the environment, has also documented increasing numbers of blacklegged ticks across NYS, including expansion into northern and western regions.[10] The New York State Department of Health (NYSDOH) has mapped the regional risk of encountering a tick infected with a disease-causing pathogen across the state based on the estimated prevalence of infected adult ticks in different geographic regions (Figure 1).[11] In this map, darker colors represent areas with a higher likelihood of encountering an infected tick. As shown in *Figure 1*, the Finger Lakes region is categorized as a very high-risk area for encountering ticks capable of transmitting disease.[11] If prevention, surveillance and risk communication efforts remain unchanged, New York State may continue to experience rising tick-born disease incidence and widening disparities in exposure and diagnosis.

Figure 1. Regional Risk of Encountering Infected Blacklegged (Deer) Ticks in NYS



(NYDOH, *Tick Risk Score (0-5 Scale) by region*, 2025) [11]

Regional Context: The Finger Lakes of New York State

The Finger Lakes region of NYS provides an important local context for addressing tick-borne disease risk due to its geographic, demographic, and health system characteristics. The region, located in western New York State, includes nine counties- Chemung, Livingston, Monroe, Ontario, Schuyler, Seneca, Steuben, Wayne and Yates- and contains a mix of urban centers and predominantly rural communities.[12,13] According to the Finger Lakes Regional Community Health Assessment (CHA), the region is home to approximately 1.28 million residents and is projected to experience a growing Environmental features of this region such as forested landscapes, agricultural land and outdoor recreation areas (i.e. lakes and trails) support tick habitats and increase the opportunities for human exposure through activities such as hiking, farming, and vineyard work. [13,14] In addition, the regional CHA identifies transportation barriers and financial constraints as key challenges that can further limit residents' abilities to access timely healthcare and health education, particularly in these rural areas.[12,13] Together ecological, environmental and physical conditions contribute to increased vulnerability to tick-borne disease exposure in the Finger Lakes region and highlight the need for targeted public health strategies.

Local Populations at Elevated Risk and Exposure Factors

In the Finger Lakes region, tickborne disease risk is not evenly distributed across populations but shaped by population-level exposure patterns and structural barriers. In addition to the high proportion of higher-risk individuals based on disease etiology local surveillance data show that incidence rates are highest among children aged 0-10 years and adults aged 40-60 years, with males consistently demonstrating higher reported case rates.[15] These patterns likely reflect the differences in occupational and recreational exposure to outdoor environments.[16,17] Outdoor workers such as forestry workers and agricultural laborers experience significantly higher rates of tick encounters and prior tickborne disease diagnosis than indoor workers.[18,19] Pet ownership can also increase exposure risk because dogs and outdoor animals may transport ticks into residential environments.[18] In the Finger Lakes region, these exposure risks may be compounded by structural barriers, therefore requiring targeted and equitably-driven interventions.

Existing Prevention Programs and Resources

Across the region, multiple organizations collaborate to reduce the risk and incidence of tick-borne diseases (TBD). For example, county health departments play an important role by monitoring reported Lyme disease cases, conducting local surveillance and providing prevention education to community members through public health advisories, online resources and outreach campaigns.[28] Local healthcare systems, including Cayuga Health and Centralus Health also contribute to prevention and early detection efforts by providing clinical guidance, diagnosing tick-borne infections, and educating patients on tick-bite prevention or symptom recognition.[29,32] In addition, organizations such as Cornell Cooperative Extension support community-based education by sharing regionally relevant information about tick identification, prevention, and safe outdoor practice – through means by workshops, outreach and online-based resources.[30,31] Lastly, these local efforts are further supported by state-level initiatives led by the New York State Department of Health, which conducts surveillance, publishes regional risk assessments (RRA) and provides educational materials to help communities understand their risks.[6,11-13]

Several educational tools have also been developed to increase public awareness and engagement. The Tick App, developed by Cornell Cooperative Extension, the University of Wisconsin-Madison, and Columbia University, is designed for the general public and includes photo identification features and a platform where users can log tick encounters.[24] This tool supports both citizen science and public awareness by helping track tick activity across New York State. In addition, Cornell Integrated Pest Management (IPM) developed the "Don't Get Ticked New York" campaign to improve community knowledge about tick prevention and tick-borne diseases.[25] While the campaign provides easy-to-understand materials and practical prevention guidance, communication with community partner Joellen Lampman suggests that gaps in engagement and public understanding still remain.[25] Similarly, the New York State Department of Health developed the "Be Tick Free" campaign to promote Lyme disease awareness and prevention behaviors.[26] Although the materials are written in clear and accessible language, the campaign focuses primarily on Lyme disease and does not address other tick-borne infections that are also present in the region.

Across all three interventions, one potential way to improve reach and effectiveness is to increase the perceived risk of acquiring an infection from a tick-bite. A study conducted by the CDC highlights gaps in public engagement with tick-prevention information. In this study, only 39% of community members reported seeing educational materials related to tick-prevention, such as posters or flyers.[27] Furthermore, among individuals who reported seeing educational materials, 75% stated performing tick-checks following outdoor activities.[27] In comparison, only 64% of individuals who had not seen educational materials reported performing tick-checks.[27] These findings suggest that increased exposure to educational materials may be associated with greater adoption of tick-prevention behaviors. By increasing awareness and improving access to prevention of resources already available in the community, similar improvements in protective behaviors may be possible.

Opportunity for a Targeted Public Health Action

Tick-borne diseases are increasing incidence and expanding northward across New York State due to climate-driven ecological shifts and inconsistent prevention behaviors.[5,7] Rural and northern communities, including the Finger Lakes region, face growing exposure risk. Current prevention strategies in this region largely focus on surveillance and passive information dissemination. For example, state and local health departments monitor tick populations and disease incidence and provide educational materials to inform communities about prevention strategies such as tick checks, protective clothing, and repellent use. However, existing literature suggests that surveillance and poorly distributed, passive information dissemination alone may not be sufficient to change prevention behaviors at the population level. We aim to address this gap.

Community Asset Summary

The Finger Lakes region has a robust ecosystem of partners whose existing infrastructure, expertise, and community trust the TickSafe Initiative will leverage and strengthen. Key stakeholder Joellen Lampman (Cornell IPM) has confirmed ongoing gaps in public engagement with existing resources.[25] The TickSafe Initiative fills the bridge between these established assets and community adoption which activates existing infrastructure for active, population-level prevention.

Public Health Focus Area	Tick-borne disease: Lyme disease and other illnesses transmitted by Ixodes scapularis
Geographic Location	Finger Lakes region of New York State
Need To Be Met	Improve prevention behaviors, awareness, and early recognition related to tick-borne diseases to decrease incidence of tick-borne disease
Key Populations to Serve	Children, adults ages 40-60, outdoor workers, rural residents, pet owners, and people frequently engaged in outdoor recreation or agricultural work
Barriers to Access	Inconsistent prevention behaviors limited public engagement with current messaging, transportation barriers, financial constraints, rural healthcare access issues, underdiagnosis, and underreporting.[7,19,22]
Facilitators of Success	Existing NYS surveillance efforts, Cornell-based outreach and education tools, local public health infrastructure, and strong community relevance due to widespread outdoor activity.[3,21,22,25]
State Health Dept. Focus; Assets to Support Success	The New York State Department of Health supports tick-borne disease prevention through surveillance reporting and educational materials such as Be Tick Free.[13,25] These resources can support local prevention messaging, risk communication, and public education.
Other State Gov. Focus; Assets to Support Success	Cornell Integrated Pest Management and Cornell Cooperative Extension provide education, outreach, and public-facing prevention resources, including Don't Get Ticked New York and promotion of The Tick App.[21,22] These state-connected resources can support locally tailored education and community engagement.
Local Health Dept. Focus; Assets to Support Success	Local and regional health departments in the Finger Lakes can use NYS surveillance data, prevention materials, and regional health assessment findings to support outreach, education, and referral.[13,19]
Other Local Gov. Focus; Assets to Support Success	County and municipal parks, recreation systems, schools, and land-use agencies may help support signage, environmental risk reduction, and public information in high-use outdoor spaces where tick exposure occurs.[17,19,20]
Local Health Care Focus; Assets to Support Success	Local healthcare systems are important for diagnosis, treatment, patient education, and reporting of tick-borne disease cases. However, access barriers suggest room for improved provider awareness and early recognition.
Local Comm. Org. Focus; Assets to Support Success	Cornell-affiliated partners, community educators, and regional health assessment networks already contribute to education and awareness efforts.[19,21,22] These organizations can help strengthen outreach, culturally relevant messaging, and local engagement.

Project Description

Program Description and Implementation Strategy

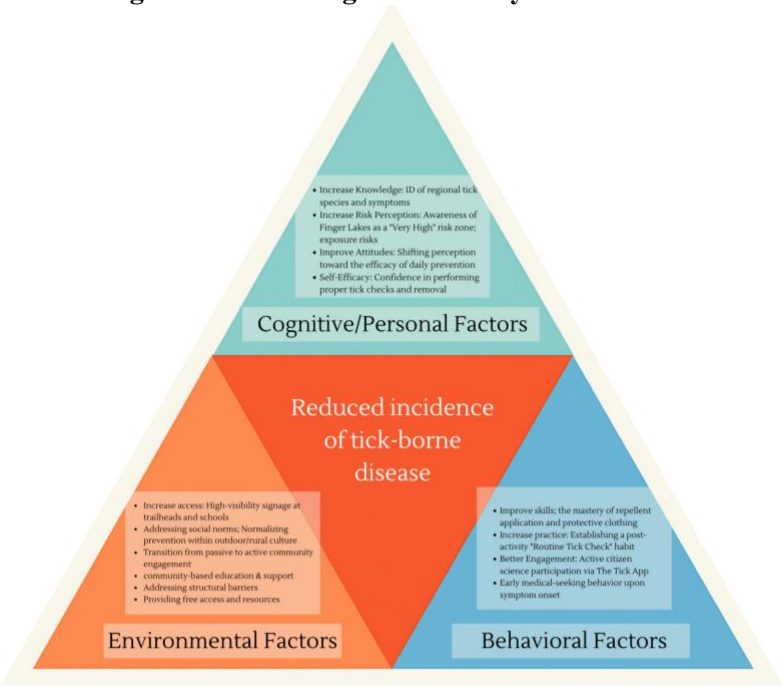
Despite increasing awareness of tick-borne diseases and the availability of prevention resources, significant gaps remain in public knowledge, risk perception, and consistent adoption of protective behaviors. In the Finger Lakes region, rising tick populations and expanding geographic risk are not matched by equivalent increases in community engagement with prevention strategies.[6,12,13] However, in order to achieve these long-term outcomes, existing gaps in awareness and community Many individuals remain unaware of their personal risk, underestimate the likelihood of exposure, or do not consistently perform recommended behaviors such as tick checks or repellent use.[27] However, in order to achieve these long-term outcomes, existing gaps in awareness and community Additionally, existing prevention efforts are often limited to passive information dissemination, which may not effectively reach high-risk populations or translate into sustained behavior change.[6,12,13,27] However, in order to achieve these long-term outcomes, existing gaps in awareness and community These persistent gaps highlight the need for more targeted, accessible, and engaging public health approaches to improve prevention and reduce disease burden.

With the northern migration of ticks into the Finger Lakes Region of NY, there are clear opportunities to improve programming and systems to reduce the morbidity and mortality of tick-borne diseases. **The overarching focus of our project is to reduce the incidence of tick-borne diseases in the Finger Lakes region by improvement of public understanding of tick exposure risk and strengthening the adoption of protective behaviors that includes performing routine tick checks, using effective repellents (e.g. DEET or permethrin-treated clothing) and recognizing early symptoms of disease.[1,2]** However, in order to achieve these long-term outcomes, existing gaps in awareness and community engagement must first be addressed. Current prevention efforts in the region largely focus on surveillance and passive information dissemination, which may not translate into sustained behavior change among community members.[6,12,13]Current research suggests that passive information dissemination alone may not be sufficient to change prevention behaviors at the population level.

Theory of Change: Applying Social Cognitive Theory to Tick-Prevention Behavior

To guide our intervention, we draw on the Social Cognitive Theory which suggests that health behaviors are shaped through the interaction of personal factors, environmental influences, and behavioral skills.[33] As depicted in *Figure 2 (Social Cognitive Theory)*, in the absence of a targeted intervention, gaps in knowledge, limited visibility of prevention messaging, and low perceived risk contribute to inconsistent adoption of tick-prevention behaviors among community members.[33] To close these gaps, our intervention will focus on increasing awareness of disease risk while encouraging consistent prevention behaviors. Guided by this framework, our proposed intervention will strengthen personal factors by increasing knowledge and awareness of tick-borne disease risk through targeted educational outreach. Environmental influences will be addressed through partnerships with local organizations to deliver visible and locally relevant prevention messaging in community settings where outdoor exposure frequently occurs or where community members are most reached.[12,13] Lastly, our intervention will promote behavioral skills by providing clear guidance on performing tick checks, using repellents correctly, and identifying early symptoms of illness[1,2] - thereby helping individuals build the confidence and self-efficacy needed to adopt protective behaviors.[33]

Figure 2. Social Cognitive Theory Framework



Project Goals and Objectives

As a step toward achieving this long-term impact, this project year we will achieve the following goal: **increase awareness of tick-borne disease risk and adoption of prevention behaviors among the selected high-risk populations in the Finger Lakes region through a locally tailored community education initiative.** Specific objectives we will achieve during this period include (i) increase exposure to accessible tick-borne disease prevention messaging among priority populations, (ii) improve knowledge and self-reported prevention behaviors related to tick-borne disease exposure, and (iii) improve regional collaboration among community organizations, stakeholders, healthcare providers, and public health agencies to sustain prevention efforts beyond the pilot year.

The TickSafe Finger Lakes Initiative

To achieve these objectives, we propose an innovative one-year pilot project: *the* TickSafe Finger Lakes *Initiative,* a community based educational outreach program designed to increase awareness of tick-borne disease risk and promote consistent prevention behaviors among residents. This initiative will target communities at elevated risk of tick exposure and aim to reduce the incidence of tick-borne diseases; specifically Lyme disease, anaplasmosis, and babesiosis; by strengthening knowledge of risk and disease prevention, early recognition of symptoms, and knowing which actions to take. This program will be built on the existing resources that were developed by the New York State Department of Health and Cornell Integrated Pest Management. It will revise and expand current educational materials to make them more engaging, accessible, and locally relevant.[13,21,25] The program will aim to create an exposure-to-prevention messaging system in the spaces where residents frequently engage in outdoor activities such as doing sports, gatherings, etc. With an improved and accessible educational resources with the increased perceived relevance of tick-borne disease risk, this program aims to encourage consistent prevention behaviors such as performed tick checks, repellent use, and seek of medical care after an exposure.[1, 2, 27] As a result, the initiative aims to reduce the prevalence of tick bites and ultimately lower the incidence of tick-borne diseases within the region.

Progress in achieving our goal will be noted as we work towards three key objectives:

Objective 1: Expanding Access to Tick-Borne Disease Prevention Education: To achieve Objective (i), we will develop and disseminate revised educational materials that improve the clarity, accessibility, and visual engagement of existing tick-borne disease prevention resources. These materials will build on existing educational campaigns such as Be Tick Free, Don't Get Ticked New York, and the Tick App, which provide evidence-based information about tick-borne diseases and prevention strategies.[13,21,24,25] Educational content will be adapted to include clear visual aids, simplified language, and QR codes to link to additional online resources. Materials will emphasize prevention strategies such as wearing of protective clothing, the use of EPA-approved insect repellents, perform the tick checks after outdoor activities, and recognize early symptoms of tick-borne disease.[1,2] Outreach materials will be distributed through printed flyers, posters, and digital media platforms to ensure accessibility across both rural and urban populations. Evidence suggests that individuals who encounter educational materials about tick prevention are more likely to engage in protective behaviors, which highlights the importance of increasing exposure to prevention messaging within communities.[27] We anticipate that increased access to easy-to-understand educational materials will improve awareness of tick-borne disease risks and encourage preventive behaviors.

Objective 2: Community-Based Outreach to Increase Prevention Engagement: To achieve Objective (ii), we will conduct community-based outreach activities designed to increase engagement with prevention messaging among populations at elevated risk of tick exposure. Target populations will include children and parents, adults aged 40–60 years, outdoor workers, rural residents, and individuals who frequently participate in outdoor recreation such as hiking, farming, or vineyard work.[15-18] These populations experience increased risk due to occupational and recreational exposure to tick habitats.[16-18] Educational outreach will occur in community locations where residents are most likely to encounter tick exposure risk, including schools, community centers, churches, libraries, healthcare facilities, parks, playgrounds, and trailheads.[17,19] Workshops and informational events will provide opportunities for residents to learn about prevention strategies, ask questions, and receive guidance on tick removal and symptom recognition. Social media campaigns will also be used to reinforce prevention of messaging and normalize protective behaviors such as performing routine tick checks following outdoor activities. By delivering prevention messaging through trusted community settings and multiple communication channels, we expect to improve knowledge, increase perceived risk of tick exposure, and build consistent prevention practices.

Objective 3: Strengthening Regional Partnerships for Sustainable Prevention: To achieve objective (iii), we will empower the partnership among local health departments, healthcare providers, Cornell Cooperative Extension programs, and community organizations to support sustained tick-borne disease prevention efforts in the region. These partners already play important roles in surveillance, education, and outreach related to both tick-borne diseases and the population as a whole across the New York State.[13, 21, 25] Through collaboration with these stakeholders, the TickSafe Finger Lakes Initiative will provide educational opportunities for healthcare providers so that they can improve the recognition and diagnosis of tick-borne diseases among residents. They are the ones who are critical in diagnosing and reporting the cases and also providing the patients with the resources to educate them on how to prevent the tick-borne diseases.[1,19] Collaboration between healthcare systems and community organizations will give this project a chance to broaden its regional capacity and provide accurate risk communication, better reporting, and early diagnosis of tick-borne diseases alongside sustained prevention efforts beyond the initial pilot period.

Work Plan and Timeline

TickSafe Initiative Implementation Strategy

The *TickSafe Finger Lakes Initiative* is a one-year pilot program designed to reduce the regional burden of tick-borne diseases (TBD) by transitioning from existing passive information dissemination to an active, theory-driven outreach model. Grounded in Social Cognitive Theory, the initiative will address the interplay of personal knowledge, environmental influence, and behavioral skills to promote sustained adoption of tick-prevention behaviors among high-risk populations in the Finger Lakes region.

This project will be implemented over the course of a 12-month period through four phases:

Phase 1: Planning and Partnership Development (Months 1-2)

During the initial phase, the project team will establish and strengthen partnerships with key stakeholders. This includes local health departments, healthcare providers, Cornell Cooperative Extension, and community-based organizations. The Project Lead will coordinate stakeholder meetings to align project goals with community priorities and ensure that intervention is responsive to the regional needs. At this stage, the research team will identify priority outreach locations based on existing surveillance data and community partner inputs. These locations will include high-risk settings such as parks, schools, healthcare facilities, and recreational areas where tick exposure is more likely to occur. Clear roles and responsibilities will be defined among team members (research assistants and community partners) to support the project implementation efficiently. The resources utilized in this phase include existing New York State surveillance data, community/regional health assessments, and partner expertise. Progress will be measured through the number of stakeholder meetings conducted, partnerships established, and the finalized implementation plan.

Phase 2: Development of Educational Materials (Months 3-4)

In Phase 2, the project team will develop and adapt educational materials to improve accessibility, clarity and relevance of tick-borne disease prevention messaging. These materials will build upon existing resources such as the *Be Tick Free* campaign and Cornell Integrated Pest Management tools, while incorporating enhancements such as simplified language, visual aids, accessibility changes and QR codes linking to additional resources and applications (i.e. The Tick App). The project team will collaborate with community partners to ensure that materials are culturally appropriate and tailored to the needs of the priority populations. The material content will emphasize key prevention behaviors, such as performing routing tick checks, using EPA-approved repellents, wearing protective clothing, and recognizing early symptoms of tick-borne diseases. The resources required for this phase include existing educational materials, design platforms (i.e. Canva, Adobe), printing and distribution of supplies, and public health literature. The progress will be measured by the number of materials developed, paired with the stakeholder feedback obtained, and final approval of materials for dissemination.

Phase 3: Community-Based Implementation (Months 5-10)

The core of the TickSafe intervention will occur during Phase 3, where educational materials and outreach strategies will be implemented across the Finger Lakes region. Prevention messaging will be disseminated through both physical and digital channels – including posters, flyers, pamphlets and social media campaigns. The community-based outreach activities will include workshops and informational sessions conducted in accessible and trusted settings such as schools, community centers, libraries, nature centers, parks, and healthcare facilities. These sessions will provide opportunities for participants to learn practical prevention skills (i.e. proper tick removal, repellent use, symptom recognition, etc.). Trained facilitators and community partners will lead to these outreach efforts. Moreover, healthcare providers will also be engaged and informed to help reinforce prevention, messaging, and improve early detection through patient education - specifically targeting parents and caregivers of children. Key resources in this phase include the physical educational materials, community spaces, trained personnel, and digital communication platforms.

The progress will be measured through outputs such as the number of outreach events conducted, the number of participants reached, the materials distributed, and social media engagement metrics. Short-term outcomes expected during this phase include increased awareness of TBD risk, improved knowledge of prevention strategies, and increased perception of susceptibility to tick exposure. These outcomes will align with the Social Cognitive Theory by strengthening both personal and environmental determinants of behavior.

Phase 4: Evaluation and Sustainability Planning (Months 11-12)
The final phase will focus on evaluating the effectiveness of the intervention as well as planning for long-term sustainability. Evaluation will be conducted using both quantitative and qualitative methods – pre and post intervention surveys - to assess changes in knowledge, attitudes and self-reported prevention behaviors. The research team will analyze collected data to determine whether the intervention was effective and reached its intended outcomes. The findings will be compiled into a final evaluation report and shared with stakeholders through dissemination meetings. The resources for this phase include survey tools, data analysis software, and stakeholder engagements meetings. The progress will be measured through completion of evaluation reports, findings, and a final sustainability plan that will outline potential program continuation or expansion. Lastly, the long-term outcomes of this initiative include increased adoption of protective behaviors, improved early detections of TBDs, and strengthened regional partnerships to support the ongoing prevention efforts. By shifting to an active, community-based educational approach, the *TickSafe Finger Lakes Initiative* aims to create sustainable improvements in the public health outcomes related to tick-borne disease prevention.

Timeline Summary

Months 1-2: Partnership development and planning
Stakeholder engagement, site selection, and program design --> Established partnerships, defined roles, and finalized implementation plan
Months 3-4: Educational Material Development
Development and adaptation of prevention materials, stakeholder feedback, and design refinement --> finalized, accessible, and locally tailored educational resources
Months 5-10: Community-based implementation
Outreach events, workshops, material dissemination, and social media campaigns --> increased reach and engagement --> improved knowledge, risk perception, and prevention behaviors
Months 11-12: Evaluation and Sustainability
Data collection, analysis, and stakeholder dissemination --> evidence of program effectiveness --> sustainability plan and long-term prevention

The full Work Plan (landscape, with all activities, outputs, responsible parties, and measures by objective) is referenced here and presented in Appendix ii.

Budget

The total budget request for this grant is $14,313.00, complemented by $5,600.00 in in-kind contributions, for a total project cost of $19,913.00. No indirect costs are included; all expenses are directly tied to program activities. The following narrative describes the assumptions and estimating parameters used to develop projected expenditures for the TickSafe Finger Lakes initiative, which is a one-year community-based tick-bite prevention program in order to serve Tompkins County, NY. Hourly wage estimates are consistent with Cornell University's student job and wage classification system. Material cost projections are based on vendor pricing and from local print suppliers and bulk supply retailers.

For a full itemized breakdown of costs, please refer to the detailed budget sheet linked here.

Budget Category	Grant Request	In-Kind	Total
Personnel	$8,100.00	$5,400.00	$13,500.00
Travel	$197.00	$0.00	$197.00
Supplies and Materials	$5,416.00	$0.00	$5,416.00
Equipment	$250.00	$0.00	$250.00
Other Expenses	$350.00	$200.00	$550.00
Total Project Costs	**$14,313.00**	**$5,600.00**	**$19,913.00**

Personnel Salaries (Salaries, Wages, and Fringe Benefits): No fringe benefits are included. The program team consists entirely of graduate students and volunteer subject-matter experts; no salaried positions are charged to this grant.

Paid Implementation (MPH Students): Three MPH students will serve as the core implementation team, responsible for all direct program delivery including event setup, community outreach, tick kit distribution, participant engagement, and on-site data collection. Hana will co-lead tabling event coordination and serve as the developer of the R Shiny web application. Asma will co-lead digital outreach and Shiny App development and help Leslie to coordinate with the Cornell IPM and NYSDOH digital partners. Leslie, alongside the management of digital coordination, will also lead community site coordination, including scheduling and logistics for library, senior center, and trailhead outreach. Each student will be compensated for 5 hours/week × $27.00/hr × 20 weeks = $2,700 per student. Total grant request: 3 students × $2,700 = $8,100.

In-Kind; Program Development (MPH Students): Each student contributes an additional 3 hrs/week × 20 weeks × $21.00/hr. = $1,260 per student ($3,780 total) in unpaid program development time, including educational material design, partner communication, website coordination, and weekly team meetings. Supported through the MPH Communications Course at Cornell University.

In-Kind; Veterinary Student: A Cornell veterinary student will volunteer subject-matter expertise on zoonotic disease, tick biology, and animal-to-human transmission at community outreach events, strengthening the program's scientific credibility with participants. 2 hrs./event × 10 events × $21.00/hr = $420 in-kind.

In-Kind; Entomology Graduate Student: A Cornell entomology graduate student will volunteer expertise on tick species identification, seasonal behavior, and habitat distribution, contributing to the

scientific accuracy of all educational materials and event programming. 2 hrs./event × 10 events × $21.00/hr. = $420 in-kind.

Travel (Transportation, Lodging, Meals, and Incidentals)

Mileage Reimbursement: Two MPH students will be reimbursed for driving outreach events across Tompkins County at the IRS standard mileage rate. 150 estimated miles/driver × $0.655/mile × 2 drivers = $197.00. No overnight travel, lodging, meals, or incidentals are anticipated, as all sites are within local driving distance.

Supplies and Materials: Materials represent the core of this intervention. Cost projections are based on vendor pricing from local print suppliers and bulk online retailers; no single-source justification is required as all items are commercially available for commodities.

Tick Kits: The program will produce and distribute 1,000 Tick Kits to Tompkins County residents at tabling and community events. Each kit contains: business cards ($90.00 × 4 packs = $360.00), plastic bags ($225.00 × 2 = $450.00), DEET wipes ($115.00 × 10 = $1,150.00), tick-checklists ($136.00 × 1 = $136.00), tweezers ($30.00 × 10 = $300.00), alcohol swabs ($40.00 × 1 = $40.00), and plastic bags to place tick in ($10.00 × 1 = $10.00).

Event and Community Materials: Printed and laminated materials support tabling events (Ithaca Farmers Market, Candor Daffodil Festival) and community education at libraries, senior centers, children's summer programs, and trailheads. Laminated flyers are specified for outdoor trailhead placement to withstand weathering. Items include: flyers ($220.00 × 1 = $220.00), activity sheets ($0.80 × 1,500 = $1,200.00), laminated flyers ($3.00 × 50 = $150.00), pens/sharpies for MPH use ($15.00 × 12 = $180.00), pens for intervention use ($10.00 × 2 = $20.00), markers ($20.00 × 2 = $40.00), poster paper ($40.00 × 3 = $120.00), crayons ($3.00 × 40 = $120.00), and stickers ($46.00 × 20 = $920.00). Total supplies and materials: $5,416.00.

Equipment

Event Equipment: A folding table ($90.00 × 1) will be used at tabling events. Three staff T-shirts ($40.00 × 3 = $120.00) provide visual identification for the outreach team. Two tick mascot plushies ($20.00 × 2 = $40.00) serve as an engagement tool for children and draw attention to tabling events. Total equipment: $250.00.

No-Cost Resources

Microsoft Office and Google Workspace are provided through Cornell University for student accounts and will be used for data storage, survey administration (Google Forms), and document management at no cost to the grant. The R Shiny web application will be hosted free on ShinyApps.io and will link directly to the Cornell IPM "Don't Get Ticked New York" campaign, the Tick App, and NYSDOH "Be Tick Free" materials. QR codes linking to these resources will be generated by the usage of free online tools.

Total Indirect Costs

No indirect costs are included in this budget. All expenses are direct to program costs.

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Appendix i: TickSafe Finger Lakes Initiative; Logic Model

Our Vision: A well-informed community protected from tick bites and tick-borne diseases					
Our INPUTS	**Our ACTIVITIES**	**Our OUTPUTS**	**Our OUTCOMES**		
What we'll invest	**What we'll do**	**What we'll create**	**Short-term results**	**Intermediate results**	**Long-term results**
~ Staff: Public health staff, community educators, health dept. personnel ~ Local/State Health Partners (health departments, Cornell Extension, NYSDOH, Cayuga Health, Parks & Rec.) ~ Community Partners (schools, parks, hardware stores, libraries, community/worship centers, etc.) ~ Funding for a one-year pilot program ~ Materials: papers, posters, pamphlets ~ Existing educational campaigns (Don't Get Ticked, Be Tick Free, etc.) ~ Data and surveillance reports ~ Technology (The Tick App)	~ Develop materials & adapt existing NYS resources into engaging, culturally relevant, visual formats (QR codes for app) ~ Conduct workshops and informational events at schools, community centers, libraries and nature centers ~ Distribute posters/flyers in high-traffic outdoor areas, and bulletin boards ~ Partner training with healthcare providers to improve recognition and awareness for caregivers ~ Place prevention messaging in parks and agricultural settings at high exposure spots	~ New sets of brochures, pamphlets, posters and digital, social media content ~ A scheduled series of community training sessions ~ A regional collaboration framework between health departments and local organizations ~ Tracking Data from Tick App citizen science; and attendee counts for outreach events	~ Increased awareness of tick-borne disease risk in the Finger Lakes ~ Improved knowledge of tick checks and early symptoms recognition ~ Enhanced self-efficacy in using repellents, removing ticks correctly ~ Positive attitude shifts regarding the necessity of prevention	~ Consistent use of protective behaviors (DEET, permethrin, protective clothing) ~ Increased frequency of routine tick checks after outdoor activities and outdoor work ~ Early medical seeking behavior after bite or symptoms ~ Strengthened collaboration between regional healthcare networks, civic spaces and public health agencies	~ Decreased incidence of Lyme Disease and other tick-borne illnesses in the Finger Lakes ~ Reduced morbidity and mortality related to VBD ~ Sustained adoption of evidence-based prevention practices (community-level) ~ Improved regional health outcomes ~ Reduced strain on the health systems

Appendix ii: TickSafe Finger Lakes Initiative; Work Plan

Goal One: Increase awareness of tick-borne disease risk and adoption of prevention behaviors among high-risk populations in the Finger Lakes region.			
Objective 1: Increase exposure to accessible tick-borne disease prevention messaging.			
Proposed Outcomes - Measures of Effectiveness • Greater awareness • Improved knowledge • Stronger visibility of prevention information			
Timeline	**Proposed Activity**	**Process of Implementation**	**Desired Output**
Month 1-2	Review/adapt existing NYS and Cornell materials	Assess gaps, simplify content, add visuals/QR codes	New brochures, posters, pamphlets, digital materials
Month 2-3	Finalize outreach materials	Pilot test with partners and revise	Ready-to-use educational materials
Month 3-12	Distribute materials across community settings	Post in parks, schools, clinics, libraries, railroads, agricultural settings...	Regional distribution of prevention materials
Objective 2: improve knowledge, self-efficacy, and prevention behaviors among priority populations			
Proposed Outcomes - Measures of Effectiveness • Good understanding of tick checks • Symptom recognition • Repellent use and stronger prevention behaviors			
Timeline	**Proposed Activity**	**Process of Implementation**	**Desired Output**
Month 3-4	Schedule community events	Coordinate with schools, parks, community centers	Outreach calendar and event schedule
Month 4-11	Conduct workshops and outreach events	Teach prevention, tick removal, and early symptoms recognition	Training sessions and attendance records
Month 5-12	Evaluate participating learning	Use short pre/post surveys and feedback forms	Evaluation data and summary report
Objective 3: Strengthen original partnership to support long-term prevention efforts			
Proposed Outcomes - Measures of Effectiveness • Stronger collaboration • Improved provider awareness • Sustainability planning			
Timeline	**Proposed Activity**	**Process of Implementation**	**Desired Output**
Month 1-2	Recruit partners	Engage health departments, Cornell extension, providers, schools, parks, community groups	Partnership group formed
Month 2-6	Hold planning meetings and create collaboration framework	Define roles and shared outreach plan	Meeting notes and collaboration framework
Month 4-10	Conduct partner/provider training	Train providers and community partners on recognition and prevention messaging	Training materials and sessions completed
Month 10-12	Sustainability planning	Review progress and identify continuation steps	Final report and sustainability plan